

Nykaa Fragrance Automation Testing Project

*Combined Selenium with Python and BDD Automation Framework
Documentation*

Synopsis

The Nykaa Fragrance Automation Testing Project is developed to automate the functionalities of the Nykaa E-commerce Fragrance module using Selenium WebDriver with Python and BDD framework concepts. The framework automates complete user workflows including homepage validation, product search, sort and filter operations, product selection, add to bag functionality, proceed operation, screenshot capture, logging mechanism, and Allure report generation.

The project demonstrates real-world QA automation testing practices by implementing modular automation architecture, reusable automation components, Page Object Model (POM), assertion validations, parameterized testing, logging, screenshot capturing, and professional reporting.

The framework supports both Selenium with Python automation and Behavior Driven Development (BDD) concepts using Behave. The project validates major E-commerce functionalities and provides scalable automation architecture suitable for modern web application testing.

Project Title

Selenium Framework

Automation Testing of Nykaa Fragrance Module Using Selenium with Python

BDD Framework

Nykaa Fragrance Automation Framework using Selenium, Python, Behave BDD, and Allure Reporting

Project Description

Selenium Framework

This project automates the Nykaa Fragrance section using Selenium WebDriver with Python. The framework validates major functionalities of the Nykaa E-commerce platform and performs end-to-end workflow testing.

Automated operations include:

- Homepage validation
- Product search
- Sort and filter operations
- Product page validation
- Add to bag functionality
- Proceed operation
- Screenshot capture
- Log generation
- Report generation

BDD Framework

The BDD framework follows Behavior Driven Development concepts using Behave and Page Object Model architecture.

The framework automates:

- Product search
- Product selection
- Add to cart functionality
- Shopping bag validation
- Positive and negative validations
- End-to-End execution flow

Reusable step definitions, screenshots, logging, parameterized testing, and Allure reports are integrated into the framework.

Test Strategy

Selenium Framework

Testing approaches implemented:

- Positive Testing
- Negative Testing
- End-to-End Testing
- Smoke Testing
- Regression Testing
- Screenshot Capture
- Logging Mechanism
- Allure Reporting

BDD Framework

BDD framework capabilities:

- Scenario Outline execution
- Parameterized testing
- Assertion-based validation
- Dynamic wait handling
- Step definition connectivity
- Page Object Model architecture
- Reusable automation utilities

Test Environment

Selenium Framework

Operating System : Windows 11

Programming Language : Python 3.14

Automation Tool : Selenium WebDriver

Framework : PyTest
Browser : Google Chrome
IDE : PyCharm
Reporting Tool : Allure Report

BDD Framework

Operating System : Windows 10 / Windows 11
Programming Language : Python 3.x
BDD Framework : Behave
Browser : Google Chrome
Driver Manager : WebDriver Manager
Logging : Python Logging Module
Version Control : Git / GitHub

Test Execution

Selenium Framework

Execution Flow:

1. Launch Browser
2. Open Nykaa Website
3. Homepage Validation
4. Search Product
5. Sort & Filter Operations
6. Product Selection
7. Add to Bag
8. Open Shopping Bag
9. Proceed Operation
10. Screenshot Capture
11. Report Generation

BDD Framework

BDD Execution Flow:

1. Open Nykaa Website

2. Search Fragrance Product
3. Select Product
4. Validate Product Title
5. Add Product to Cart
6. Open Shopping Bag
7. Proceed Further
8. Validate End-to-End Flow

Validation Scenarios:

- 4 Positive Test Cases
- 2 Negative Test Cases

Test Output

Selenium Framework

Outputs generated:

- Browser execution logs
- Screenshot evidence
- Pass/Fail execution status
- Console output
- Allure report data

BDD Framework

Outputs generated:

- Execution results
- Assertion validations
- Allure reports
- Screenshots after every step
- Step execution logs

Test Reports

Selenium Framework

Allure Reports provide:

- Execution summary

- Passed and failed test cases
- Execution duration
- Screenshots
- Logs

BDD Framework

BDD Allure reports contain:

- Scenario execution details
- Step-by-step execution
- Screenshot attachments
- Error stack traces
- Validation reports

Conclusion

Combined Conclusion

The Nykaa Fragrance Automation Testing Project successfully demonstrates Selenium automation testing using Python with both Selenium and BDD framework concepts.

The framework includes reusable automation scripts, logging mechanisms, screenshot capture, parameterized testing, assertion validations, report generation, and complete End-to-End workflow automation.

The project reflects real-world QA automation practices and provides scalable automation framework architecture suitable for modern E-commerce application testing.